

TOTMA, Russian Federation

WMO: 270510

Lat: 59.9806N

Lon: 42.7694E

Elev: 135

StdP: 99.71

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.4	-26.8	-33.2	0.2	-30.4	-29.3	0.3	-26.8	9.3	-6.5	8.0	-7.6	1.4	10	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	28.2	19.7	26.2	18.8	24.4	17.8	21.3	26.4	19.9	24.4	18.7	22.9	2.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.4	24.0	18.2	13.3	22.3	17.0	12.3	21.0	62.5	26.2	57.4	24.5	53.6	22.9	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.0	6.0	-32.9	30.9	3.8	1.7	-35.7	32.2	-37.9	33.2	-40.0	34.1	-42.8	35.4		
(5)			-32.8	22.8	3.8	1.5	-35.6	23.9	-37.8	24.7	-40.0	25.6	-42.8	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-11.4	-10.0	-4.4	3.3	10.3	14.6	17.7	14.9	9.6	3.1	-3.6	-8.0
	DBStd	11.39	8.13	7.60	5.18	4.67	5.11	4.32	3.67	3.89	3.80	4.21	5.88	7.64
	HDD10.0	3186	664	561	447	206	61	9	1	4	52	216	407	558
	HDD18.3	5635	923	794	705	451	253	128	56	119	262	471	657	817
	CDD10.0	662	0	0	0	6	70	146	240	157	40	3	0	0
	CDD18.3	71	0	0	0	0	3	15	38	14	1	0	0	0
	CDH23.3	637	0	0	0	1	46	154	319	114	2	0	0	0
	CDH26.7	137	0	0	0	0	7	35	67	28	0	0	0	0
Wind	WSAvg	2.7	2.6	2.8	2.9	2.8	3.0	2.7	2.3	2.4	2.5	2.8	2.8	2.9
Precipitation	PrecAvg	618	40	29	31	40	45	70	75	71	60	59	45	46
	PrecMax	858	76	54	86	94	88	195	146	184	171	122	86	95
	PrecMin	426	11	1	3	1	8	12	9	7	24	0	13	14
	PrecStd	78	15	13	17	23	22	38	31	34	27	27	18	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	2.7	9.3	21.4	27.7	30.4	31.0	29.8	23.2	16.5	8.0	5.0
		MCWB	0.9	1.3	4.6	12.6	17.3	20.8	21.8	21.7	16.7	12.0	7.1	4.7
	2%	DB	1.0	1.5	5.8	17.1	24.3	27.1	28.4	26.3	20.0	12.8	5.5	2.7
		MCWB	0.3	0.5	2.1	9.8	15.7	18.5	20.4	19.0	15.0	10.4	4.8	2.0
	5%	DB	0.4	0.7	3.7	13.6	21.6	24.8	26.7	23.8	17.6	10.6	4.2	1.4
		MCWB	-0.2	-0.1	1.3	7.8	13.8	17.3	19.5	18.0	13.8	8.9	3.6	0.7
	10%	DB	-0.8	-0.3	2.3	11.0	19.2	22.7	25.0	21.6	15.7	9.0	2.7	0.5
		MCWB	-1.5	-1.0	0.4	6.5	12.5	16.4	18.6	17.1	12.9	7.8	2.1	0.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.6	5.2	13.0	18.9	21.5	23.2	23.2	17.7	12.6	7.4	4.6
		MCDB	1.4	2.3	8.2	20.7	24.4	28.2	28.3	29.3	21.5	14.8	7.9	4.8
	2%	WB	0.5	0.7	2.8	10.5	16.6	19.8	21.9	20.3	15.7	10.7	4.9	2.2
		MCDB	0.8	1.3	4.7	15.3	22.6	25.3	27.1	24.4	18.7	12.6	5.4	2.7
	5%	WB	-0.1	0.0	1.7	8.6	14.9	18.4	20.5	18.7	14.5	9.2	3.6	0.8
		MCDB	0.4	0.6	3.2	12.9	20.0	23.3	24.9	22.3	17.0	10.4	4.0	1.2
	10%	WB	-1.5	-1.1	0.7	6.8	13.3	17.0	19.3	17.6	13.1	7.8	2.2	0.0
		MCDB	-0.9	-0.3	1.9	10.6	18.2	21.5	23.7	21.0	15.3	9.0	2.6	0.4
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.9	8.2	9.4	10.4	9.8	10.0	9.0	7.5	4.6	3.8	4.7
		MCDBR	4.9	4.7	9.3	14.3	14.2	13.1	12.8	12.3	11.3	7.8	4.5	3.7
		MCWBR	4.8	4.5	6.4	8.4	7.0	6.1	5.9	6.0	6.6	5.4	4.2	3.6
	5% WB	MCDBR	4.6	4.7	6.9	13.0	12.6	11.7	11.4	10.5	9.9	7.1	4.3	3.6
		MCWBR	4.7	4.6	5.0	8.3	6.8	6.0	6.1	5.8	6.2	5.5	4.2	3.6
Clear-Sky Solar Irradiance	taub	0.262	0.304	0.333	0.376	0.360	0.356	0.391	0.380	0.347	0.327	0.286	0.240	
	taud	2.043	2.064	2.077	2.183	2.421	2.454	2.368	2.378	2.457	2.374	2.148	2.001	
	Ebn at Noon	484	666	776	815	868	876	830	810	780	662	457	357	
	Edh at Noon	47	81	110	120	103	101	108	99	78	61	42	32	
All-Sky Solar Radiation	RadAvg	0.33	0.99	2.27	3.68	5.01	5.25	5.30	3.85	2.20	0.91	0.30	0.17	
	RadStd	0.06	0.26	0.37	0.45	0.39	0.49	0.46	0.33	0.26	0.15	0.05	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page